

Problem E: Jayden's Automaton Factory

Time Limit: 5.0s | Memory Limit: 1024MB

Description

Having mastered the art of the manual T-Spin, Jayden has turned his attention to the theoretical limits of stacking. He is designing a machine to generate the "Perfect Loop"—an infinite sequence of pieces that keeps the board perfectly flat.

The game is played on a grid of width W and infinite height. The pieces are the standard 7 Tetrominoes (I, O, T, L, J, S, Z). **Pieces do not rotate.** They fall from the top of the grid in a specified column C , settling when they collide with the floor or existing blocks. If a row becomes completely full of blocks, it clears instantly, and all blocks above it shift down by 1.

However, the sequence of pieces is not random. It is determined by a **Bag Automaton**. The automaton is a directed graph with N states (numbered 1 to N). Each edge in the graph is labeled with a Piece Type P and a Column Index C . Transitioning along an edge (u, v) with label (P, C) means Jayden drops piece P at column C .

Jayden wants to know: How many distinct paths of length L exist in the automaton, starting from state 1, such that after all L pieces are dropped and all line clears are processed, the board is **completely flat** (all columns have height 0)?

Input

The first line contains three integers: W , N , and L .

$$1 \leq W \leq 6, \quad 1 \leq N \leq 50, \quad 1 \leq L \leq 10^{18}$$

The second line contains an integer M ($1 \leq M \leq 200$), the number of edges in the automaton. The next M lines describe the edges. Each line contains:

$$u \quad v \quad T \quad C$$

representing a transition from state u to state v by dropping piece type $T \in \{I', O', T', L', J', S', Z'\}$ at column C ($1 \leq C$). It is guaranteed that the piece fits horizontally within the width W at column C .

Output

Output a single integer: the number of valid paths of length L modulo $10^9 + 7$.

Examples

Example 1

Input:

```
4 1 2
2
1 1 0 1
1 1 0 3
```

Output:

```
2
```

Explanation: The board width is 4. Path 1: Drop 'O' at col 1. Occupies cols 1-2, rows 1-2. Path 2: Drop 'O' at col 3. Occupies cols 3-4, rows 1-2. After both moves, rows 1 and 2 are full and clear. The board returns to flat state (0). The reverse order (O at 3, then O at 1) is also valid. Total = 2.

Example 2

Input:

```
3 2 10000000000000000000
2
1 2 I 1
2 1 L 1
```

Output:

```
0
```